JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAM
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY

9744/04

Paper 4 Practical

15 August 2017

CCONFIDENTIAL INSTRUCTIONS

2 hours 30 minutes

Great care should be taken to ensure that any confidential information, including the identity of material on microscope slides where appropriate, does not reach the candidates either directly or indirectly.

This document consists of **7** printed pages.

[Turn over

Question 1

Preparation list

	Apparatus/Reagents/Chemicals	Quantity per student
1	2,6-dichlorophenolindophenol (DCPIP) in a specimen tube with a cap, labelled **D**.	$3 cm^3$
2	Leaf extract in a specimen tube, labelled **L**, in a beaker containing ice	$10 cm^3$
3	Graduated $1 cm^3$ Pasteur pipette	1
4	$1 cm^3$ syringes	2
5	6cm × 15cm filters, folded longitudinally to form a 'tent', of the following colours: purple, labelled **P** (approx. 425nm) blue, labelled **B** (approx. 450nm) green, labelled **G** (approx. 525nm) orange, labelled **O** (approx. 625nm) red, labelled **R** (approx. 675nm) Filters should be labelled in one corner.	1 of each colour
6	White tile – 10cm × 10cm	1
7	Length of aluminium foil, large enough to wrap around the specimen tube containing the leaf extract	1 piece
8	Square piece of aluminium foil, large enough to form a lid for the specimen tube containing the leaf extract	1 piece
9	Bench lamp with 60W filament bulb (or equivalent)	1
10	Stop clock or stopwatch	1
11	30cm ruler	1
12	Glass rod	1
13	Safety glasses/goggles	1 pair
14	$250 cm^3$ beaker containing distilled water, labelled **for washing**	1
15	$250 cm^3$ beaker, labelled **for waste**	1
16	Paper towels	5
17	Disposable gloves	1 pair

Instructions for preparation

Buffer solution, pH 7.5

4.5g disodium hydrogenphosphate ($Na_2HPO_4.12H_2O$)
1.7g potassium dihydrogenphosphate (KH_2PO_4)
500cm^3 water

Dissolve the disodium hydrogenphosphate and potassium dihydrogenphosphate in 450cm^3 distilled or deionised water.
When the solids have completely dissolved make the volume up to 500 cm^3 with distilled or deionised water.
Check the pH of the buffer and adjust if necessary. Adding potassium dihydrogenphosphate will lower pH, adding disodium hydrogenphosphate will increase pH.

2,6-dichlorophenolindophenol (DCPIP) solution, labelled **D**

0.2 g DCPIP
0.9 g potassium chloride

Dissolve in 250 cm^3 of buffer solution.
This can be made before the examination and stored in a refrigerator.
The concentration being used by candidates does not constitute a hazard.

Extraction solution

13.7 g sucrose
0.1 g potassium chloride

Dissolve in 100 cm^3 of buffer solution.
Refrigerate until needed.

Leaf extract, labelled **L** and with hazard symbol for irritant

This should be prepared on the day on which the practical is to be carried out.

Use approximately 15 g spinach leaves (or any soft, dark green leaves) for each 100 cm^3 of the extraction solution.

Use scissors to cut the midrib and any large veins from the leaves. Cut the remaining leaf material into small pieces and add to the extraction solution. Use a liquidiser or blender to separate and disrupt the cells. Filter the leaf extract through muslin or fine mesh fabric to remove leaf debris. Store the filtrate in a beaker in a refrigerator before dispensing to candidates.

This should be provided to students in a beaker of ice.

Filters

These should be obtained from a photographic supplier and should allow light of the approximate wavelengths required to pass through without large differences in overall absorption.

Lee Filters (obtainable from www.hwarta.com) provide a range of suitable filters other than the codes stated in the preparation list.

	Specified Codes	**Examples of Alternatives**
purple	136	052, 170
blue	140	144, 117
green	139	124
orange	105	021
red	164	022

Question 2

Preparation list

Each candidate must have sole, uninterrupted use of a microscope for 1 hour 15 minutes only.

For each candidate:

- the microscope must be set up on the low-power objective lens
- the slide must not be left in the stage or n the microscope

Apparatus for each candidate	**Quantity**	✓
Microscope slides and coverslips	3	
Pipette, teat to remove samples from containers	2	
Glass rod	1	
Mounted needle or seeker	1	
Beaker or container, (about 200 cm^3) with tap water, labelled **for washing**	1	
Beakers or container, labelled **for waste**	1	
Glass marker pen	1	
Paper towel	6	
Stop-clock or stopwatch or sight of a clock to time five minutes	1	
Microscope with: • Low-power objective lens, × 10 (equal to 16 mm or $\frac{2}{3}$") • High-power objective lens, × 40 (equal to 4 mm or $\frac{1}{6}$") • Eyepiece lens, × 10 (equal to 16 mm or $\frac{2}{3}$") • Eyepiece graticule fitted within the eyepiece and visible in focus at the same time as the specimen.	1	

- Yeast cell suspensions provided to the candidates should be supplied in a container, suitable for the removal of a drop of suspension using a glass rod or teat pipette.

Prepare a 7.0% yeast cell suspension. **The glucose should be added to the yeast 10 to 15 minutes before the candidates start Question 2. Each candidate should have fresh yeast cell suspension.**

As the yeast cell suspension will froth, it should be prepared in a large container.

7.0 g of dried yeast (for baking) is added to 80 cm^3 of warm distilled water, stirred and made up to 100 cm^3 with warm distilled water. This should be kept at a temperature between 35°C and 40°C.

10 to 15 minutes before the candidates start Question 2 add the glucose.

Sprinkle 20 g of glucose, a little at a time, onto the surface of the yeast cell suspension, stirring continuously. Keep warm between 35°C and 40°C until needed.

Put 50 cm^3 of the 7.0% yeast suspension (that you have prep above) into a beaker or container and make up to 100 cm^3 with warm distilled water. The yeast needs to be actively frothing.

This makes the 3.5% yeast suspension needed for the samples **S2** and **S3**.

S1 could be prepared the day before and stored in a refrigerator.

Summary of solutions and reagents:

labelled	contents	hazard	volume / cm^3
S1	3.5% boiled yeast cell suspension	none	at least 5
S2	3.5% active yeast cell suspension	none	at least 5
S3	mixture of S1 and S2	none	at least 5
M	1% methylene blue solution	[H] harmful	at least 5

It is advisable to wear safety glasses/goggles when handling chemicals.

Each candidate will require:

(i) S1, at least 5 cm^3 of 3.5% boiled yeast cell suspension in a small container, labelled **S1**. For example: put 20 cm^3 of 3.5% yeast cell suspension into a container and put this into a boiling water-bath for 10 minutes.

To check that **all** the yeast cells are dead:
- place a drop of suspension onto a microscope slide
- add 1 drop of methylene blue as made up below
- mix with a glass rod and leave for 5 minutes
- add a coverslip and observe using the high-power objective lens of the microscope.
- Nearly all the cells should be blue. If this is not the case then boil for an extra 5 minutes. Repeat if necessary.

S1 could be prepared the day before and stored in a refrigerator.

This is sufficient for 4 candidates.

(ii) S2, at least 5 cm^3 of 3.5% yeast cell suspension (active) as diluted above from active 7.0% yeast suspension.

(iii) S3, at least 5 cm^3 of 3.5% yeast cell suspension made up of equal volumes of **S1** and **S2**. For example, put 10 cm^3 of S1 into a small container. Add 10 cm^3 of **S2** to the small container and mix well.

This is sufficient for 4 candidates.

(iv) **M**, at least 5 cm^3 of freshly prepared 1% methylene blue solution in a container with a pipette, labelled **M**.

This is prepared by dissolving 1.0 g of methylene blue and 0.6 g of sodium chloride in 80 cm^3 of distilled water and making it up to 100 cm^3 with distilled water.

This is sufficient for 20 candidates.

(Safety: Be careful not to inhale the powder. If methylene blue comes into contact with your skin, rinse with cold water.)

Apparatus for each group of candidates should be clean.

Question 3

No materials are required for this question.